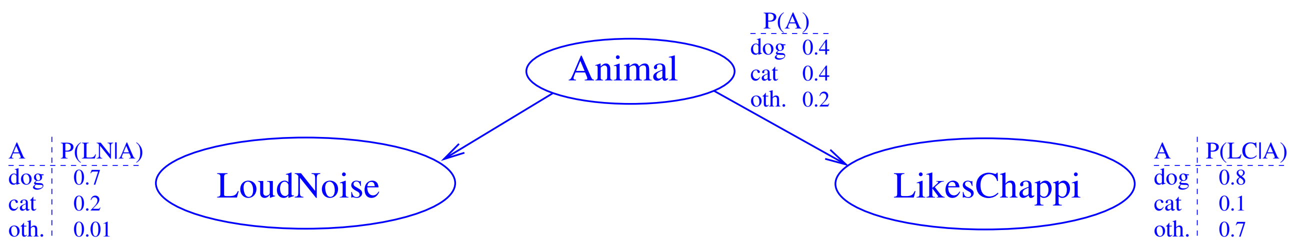
Exercise 1 :

Consider the following Bayesian network BN:



Use inference by enumeration to compute the following probabilities:

(a) $P(dog \mid loudnoise, likeschappi)$.

(b) $P(loudnoise \mid \neg likeschappi)$.

Include intermediate steps at a level of granularity as in the lecture slides examples. In particular, for each of (a) and (b), state what the query variable, evidence, and hidden variables are; and write down *which* probabilities provided in BN can be combined *how* to obtain the demanded probability P.

Solution:

As the variable ordering consistent with BN, we choose $X_1 = Animal$, $X_2 = LoudNoise$, $X_3 = LikesChappi$.

(a) The query variable X here is $Animal$. The evidence $\mathbf{e}$ is $loudnoise, likeschappi$. There are no hidden variables, $\mathbf{Y} = \emptyset$. Using Normalization+Marginalization $\mathbf{P}(X \mid \mathbf{e}) = \alpha \sum_{\mathbf{y} \in \mathbf{Y}} \mathbf{P}(X, \mathbf{e}, \mathbf{y})$ we get $\mathbf{P}(Animal \mid loudnoise, likeschappi) = \alpha \mathbf{P}(Animal, loudnoise, likeschappi)$. By the Chain rule and exploiting conditional independence, we get $\alpha \mathbf{P}(Animal, loudnoise, likeschappi) = \alpha \mathbf{P}(likeschappi \mid Animal) * \mathbf{P}(loudnoise \mid Animal) * \mathbf{P}(Animal) = \alpha \langle 0.8 * 0.7 * 0.4, 0.1 * 0.2 * 0.4, 0.7 * 0.01 * 0.2 \rangle \approx \langle 0.96, 0.03, 0.01 \rangle$. Thus $P(dog \mid loudnoise, likeschappi) \approx 0.96$. ("If your animal likes Chappi and makes loud noise, chances are good it's a dog.")

(b) The query variable X here is $LoudNoise$. The evidence $\mathbf{e}$ is $\neg likeschappi$. The hidden variables $\mathbf{Y}$ are $\{Animal\}$. Using Normalization+Marginalization $\mathbf{P}(X \mid \mathbf{e}) = \alpha \sum_{\mathbf{y} \in \mathbf{Y}} \mathbf{P}(X, \mathbf{e}, \mathbf{y})$ we get $\mathbf{P}(LoudNoise \mid \neg likeschappi) = \alpha \sum_{\mathbf{y} \in \{Animal\}} \mathbf{P}(\mathbf{y}, LoudNoise, \neg likeschappi)$. By the Chain rule and exploiting conditional independence, we get $\alpha \sum_{\mathbf{y} \in \{Animal\}} \mathbf{P}(\mathbf{y}, LoudNoise, \neg likeschappi) = \alpha \sum_{\mathbf{y} \in \{Animal\}} P(\neg likeschappi \mid \mathbf{y}) * \mathbf{P}(LoudNoise \mid \mathbf{y}) * P(\mathbf{y})$. That is, for each truth value v of $LoudNoise$, we need to sum over each possible animal $\mathbf{y}$ the product of the likelihood of $\mathbf{y}$ not liking Chappi multiplied with the likelihood of $\mathbf{y}$ having value v for $LoudNoise$ multiplied with the likelihood of $\mathbf{y}$. This gives us $\mathbf{P}(LoudNoise \mid \neg likeschappi) = \alpha \langle 0.2 * 0.7 * 0.4 + 0.9 * 0.2 * 0.4 + 0.3 * 0.01 * 0.2, 0.2 * 0.3 * 0.4 + 0.9 * 0.8 * 0.4 + 0.3 * 0.99 * 0.2 \rangle = \alpha \langle 0.1286, 0.3714 \rangle = \langle 0.2572, 0.7428 \rangle$. Thus $P(loudnoise \mid \neg likeschappi) = 0.2572$. ("If your animal does not like Chappi, chances are good it is quiet.")

Exercise 2 :

In the graphs in Figures 1 and 2, determine which variables are d-separated from A. Note that it is sufficient to find a single open path along which evidence can be transmitted; if such a path exists then the variables are *not* d-separated (instead they are said to be d-connected).

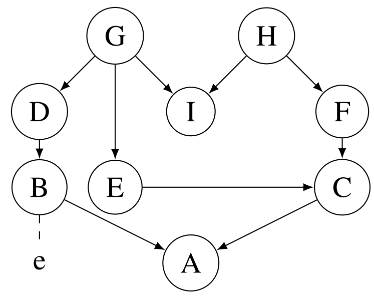


Figure 1: Figure for Exercise 2.

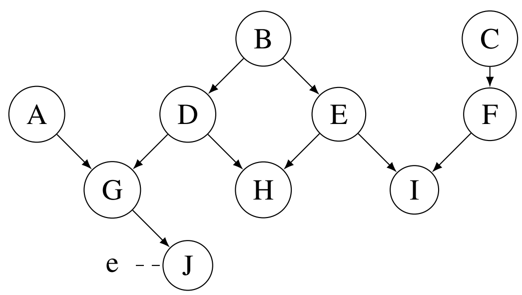


Figure 2: Figure for Exercise 2.

Solution:

In (a), all variables are d-connected to A. We can see that by checking whether there *exists* an open path between each pair of variables. Take A and D for instance. The path $A \leftarrow C \leftarrow F \leftarrow H \rightarrow I \leftarrow G \rightarrow D$ is closed, since it contains the connection ($H \rightarrow I \leftarrow G$), which is converging and there is no evidence on I (nor on any of its descendants for which it has none). However, the path $A \leftarrow C \leftarrow E \leftarrow G \rightarrow D$ is open since each triplet of nodes along the path (i.e., ($A \leftarrow C \leftarrow E$), ($C \leftarrow E \leftarrow G$), ($E \leftarrow G \rightarrow D$) defines an open connection according to the d-separation rules. With this path we have established the existence of an open path, and we therefore have that A and D are no d-separated given the evidence provided.

In (b), all variables except C and F are d-connected to A.

Exercise 3 :

Consider the network in Figure 3.

- What are the minimal set(s) of variables that we should have evidence on in order to d-separate C and E (that is, sets of variables for which no proper subset d-separates C and E)?

- What are the minimal set(s) of variables we should have evidence on in order to d-separate A and B?
- What are the maximal set(s) of variables that we can have evidence on and still d-separate C and E (that is, sets of variables for which no proper superset d-separates C and E)?
- What are the maximal set(s) of variables that we can have evidence on and still d-separate A and B?

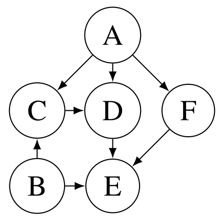


Figure 3: A causal network for Exercise **??**.

Solution:

Minimal sets d-separating C and E: $\{B, D, F\}$ and $\{A, B, D\}$.

Minimal sets d-separating A and B: $\emptyset$.

Maximal set d-separating C and E: $\{A, B, D, F\}$.

Maximal set d-separating A and B: $\{F\}$.

Exercise 4 :

Consider the network defined by the two binary variables A and B, where A is the parent of B. Assume that the conditional probability tables are given as $P(A) = (0.1, 0.9)$ and

	A	
	a_1	a_2
b_1	0.05	0.2
b_2	0.95	0.8

i) Assume that you want to estimate $P(b_1)$ using sampling. How many samples would be required if you only accept an error larger than 0.15 in 10% of the cases?

ii) Generate two random samples, using the following list of random numbers (any time that you use a random number pick one from the list): $0.5, 0.3, 0.01, 0.8$.

iii) Suppose that after sampling 10 states, you got $(A = a_2, B = b_2)$ 3 times, and $(A = a_1, B = b_2)$,6 times, and $(A = a_1, B = b_1)$ once. Estimate $P(B = b_1)$

iv) (Optional) Implement the network above in Hugin and use Hugin to sample the number of cases that you calculated in step (i); use the function 'Simulate cases' under 'File'. Use the sampled cases to estimate $P(b_1)$ and compare the result with Hugin. Feel free to use a spreadsheet for the counting.

v) Assume that you want to use rejection sampling to estimate $P(A|B = b_1)$. How many samples do you expect you would have to generate in order to end up (after rejection) with a sample set of 1000 cases for estimating the probability.

Solution:

i) Hoeffding's inequality gives us

$$P(|s-p| > 0.15) \leq 2e^{-2n0.15^2} < 0.1,$$

which we can rewrite to find that $n > 66.57$.

ii) As A is parent of B, we always sample A first.

First state: $A = a_2, B = b_2$: $0.5 > 0.1$, so $A = a_2$, then $0.3 > 0.2$ so $B = b_2$. Second state: $A = a_1, B = b_2$: $0.01 \leq 0.1$, so $A = a_1$, then $0.8 > 0.05$ so $B = b_2$.

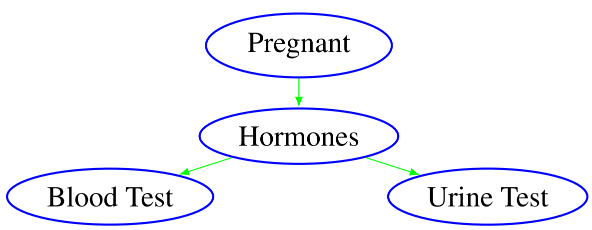
iii) $P(B = b_1) = 0.1$

iv) (Must be done in Hugin)

v) The probability of $B = b_1$ is $P(B = b_1) = 0.185$, hence only 18.5% of all the generated samples would not be rejected. We therefore need to generate approximately 5400 cases to end up with a sample set of 1000 cases.

Exercise 5 :

Consider the insemination example from Section 3.1.13 in BNDG:



Let the probabilities be as in Table 1 ($Ho = y$ means that hormonal changes have taken place) $P(Pr) = (0.87, 0.13)$.

	$Pr = y$	$Pr = n$
$Ho = y$	0.9	0.01
$Ho = n$	0.1	0.99

	$Ho = y$	$Ho = n$
$BT = y$	0.7	0.1
$BT = n$	0.3	0.9

	$Ho = y$	$Ho = n$
$UT = y$	0.8	0.1
$UT = n$	0.2	0.9

Table 1: Tables for Exercise 5.

i) What is $P(Pr \mid BT = n, UT = n)$?

ii) (Optional) Verify your solution modelling the BN in Hugin.

Solution: *Part 1:*

$$P(Pr|Bt = n, Ut = n) = \frac{P(Pr, Bt = n, Ut = n)}{P(Bt = n, Ut = n)}$$

Next, we should calculate $P(Pr, Bt = n, Ut = n)$ and $P(Bt = n, Ut = n)$:

$$P(Pr, Bt = n, Ut = n) = \sum_{Ho} P(Pr, Bt = n, Ut = n, Ho)$$
$$P(Bt = n, Ut = n) = \sum_{Pr} P(Pr, Bt = n, Ut = n)$$

Thus, we only need to calculate $P(Pr, Bt = n, Ut = n, Ho)$ and this can be done using the chain rule:

$$P(Pr, Bt = n, Ut = n, Ho) = P(Ut = n|Ho)P(Bt = n|Ho)P(Ho|Pr)P(Pr)$$

The final value is $P(Pr|Bt = n, Ut = n) = (0.53, 0.47)$. See also the Hugin network.

Exercise 6 :
Complete Exercise 9.4(a-b) in PM.

Solution:

We have two variables in the network: *Company* (C) with states *green* (g) and *blue* (b) and *Witness* (W) with states *green* (g) and *blue* (b). The structure of the network incorporating one witness is illustrated in Figure 4(a).

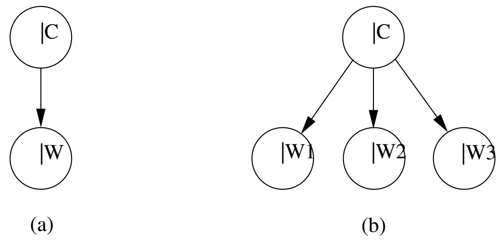


Figure 4: The witness model: Model (a) includes one witness and model (b) includes three witnesses.

The conditional probability tables for the model are given by:

$P(C=g)$	$P(C=b)$
0.85	0.15

	$C=g$	$C=b$
$W=g$	0.8	0.2
$W=b$	0.2	0.8

$P(W \mid C)$

For calculating $P(C=b \mid W=b)$ we use Bayes rule:

$$\begin{aligned} P(C=b \mid W=b) &= \frac{P(W=b \mid C=b)P(C=b)}{P(W=b)} \\ &= \frac{P(W=b \mid C=b)P(C=b)}{P(C=b, W=b) + P(C=g, W=b)} \\ &= \frac{0.8 \cdot 0.15}{0.8 \cdot 0.15 + 0.2 \cdot 0.85} \\ &= \frac{0.12}{0.12 + 0.17} \approx 0.41 \end{aligned}$$

With three independent witnesses we have the network in Figure 4(b), where we assign the same conditional probability distribution to each of the three witnesses. Thus, we have a naive Bayes model with three information variables.

We can perform probability updating in this model as shown on the slides in the previous lecture. E.g.

$$\begin{aligned} P(C=b \mid W_1=W_2=W_3=b) &= \frac{P(W_1=b \mid C=b)P(W_2=b \mid C=b)P(W_3=b \mid C=b)P(C=b)}{P(W_1=b, W_2=b, W_3=b)} \\ &= \frac{0.8 \cdot 0.8 \cdot 0.8 \cdot 0.15}{0.8 \cdot 0.8 \cdot 0.8 \cdot 0.15 + 0.2 \cdot 0.2 \cdot 0.2 \cdot 0.85} = 0.92 \end{aligned}$$

Exercise 7 :

Consider the running example of the lecture with the following probability tables:

	Burglary *t*	Burglary *f*
	.1	.9

	Earthquake *t*	Earthquake *f*
	.1	.9

Burglary	Earthquake	Alarm *t*	Alarm *f*
t	*t*	.9	.1
t	*f*	.8	.2
f	*t*	.5	.5
f	*f*	.1	.9

Alarm	JohnCalls *t*	JohnCalls *f*
t	.8	.2
f	.1	.9

Alarm	MaryCalls *t*	MaryCalls *f*
t	.7	.3
f	.1	.9

Determine the conditional probability $P(MC \mid B=t)$. For each step indicate the operations that are performed.

Solution:

Note: Below you have the solution using Variable Elimination, but the exercise can also be solved by Naive Enumeration and the result should be the same.

$$\sum_{a\in\{t,f\}}\sum_{eq\in\{t,f\}}\sum_{jc\in\{t,f\}} P(B=t)P(EQ=eq)P(A=a \mid B=t,EQ=eq)P(JC=jc \mid A=a)P(MC \mid A=a) =$$
$$\sum_{a\in\{t,f\}}\sum_{eq\in\{t,f\}} P(B=t)P(EQ=eq)P(A=a \mid B=t,EQ=eq)P(MC \mid A=a)F_1(a) =$$
$$\sum_{a\in\{t,f\}} P(B=t)P(MC \mid A=a)F_1(a)F_2(a) =$$
$$P(B=t)F_3(MC)$$

With the concrete tables we have specified for the conditional distributions, we can now compute: to eliminate the variable JohnCalls, we multiply all tables that contain the variable JohnCalls, and sum out the JohnCalls variable. The result is the table, or factor, F_1. Since there is only one table containing JohnCalls, the result is very simple:

	$F_1(Alarm)$	
	t	f
	1	1

Next we eliminate *Earthquake*. There are two tables containing Earthquake: the table of the Earthquake node, and the conditional distribution of Alarm. The latter table is restricted to the cases that are consistent with the observed evidence $B = t$, which gives a table only containing Earthquake and Alarm:

	Alarm	
Earthquake	t	f
t	.9	.1
f	.8	.2

Multiplying this with the Earthquake table gives

	Alarm	
Earthquake	t	f
t	.09	.01
f	.72	.18

Summing out the Earthquake variable then gives the factor F_2:

	$F_2(Alarm)$	
	t	f
	0.81	0.19

Finally, we eliminate *Alarm*. Multiplying F_1, F_2, and the table for MaryCalls gives

	MaryCalls	
Alarm	t	f
t	.7· 0.81=0.567	.3 · 0.81=0.243
f	.1· 0.19 = 0.019	.9 · 0.19= 0.171

Summing out Alarm gives

	$F_3(MaryCalls)$	
	t	f
	0.586	0.414

This multiplied with $0.1 = P(B = t)$ gives

	$P(B = t)F_3(MaryCalls)$	
	t	f
	0.0586	0.0414

which is now the table containing the function $P(B = t, MC)$. To obtain the conditional distribution $P(MC \mid B = t)$ this has only to be normalized by dividing with 0.0586+0.0414, which gives

	$P(MaryCalls \mid B = t)$	
	t	f
	0.586	0.414

Note that by inspecting the semantics of the potentials calculated above, we could have stopped after summing out Alarm, since the resulting potential is the conditional distribution $P(MC \mid B = t)$.